# Banking Portfolio Review

Loan risk, branch operations and growth opportunities

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## Summary

This report looks at a bank's lending and day-to-day operations through eight practical questions. Where do loans default? Do branches approve loans in a similar way? How busy are the people in each branch? Where do transactions break? How much of the loan book rests on a few large loans? And where is there room to grow? The answers come from a six-table banking dataset, analysed in SQL.

The six tables are customers (3,000 records), accounts (4,200), loans (3,200 applications), transactions (24,000), employees (120) and branches (15). Between them they cover who the customers are, the accounts and loans they hold, every transaction from January 2023 to December 2025, and the staff and branches that serve them.

On the technical side, the work uses joins across the tables (including anti-joins, which find customers who have no loan), common table expressions to build each answer step by step, and window functions to rank, compare against averages and measure change over time. Each question is also saved as a reusable database view, so the numbers can be refreshed when new data arrives.

### The five things that stand out

- About 7.7% of approved loans have defaulted, worth roughly 170 million. Education is the riskiest product, and Premium customers with Education or Auto loans show the highest default rates, though on small numbers. Mass customers account for two thirds of all defaults.
- Approval rates are fairly even across the bank, at about 75%. Two branches, Kolkata Central and Bhubaneswar Main, sit well below the rest.
- The loan book leans heavily on Home and Business lending. The largest 10% of active loans hold more than a quarter of the money, and every one of them is a Home or Business loan.
- About 3.6% of transactions fail or stay pending, which leaves roughly 12.7 million in limbo. Nagpur City is the clear problem branch.
- Most accounts marked "Active" are not being used. Nearly two in three had no completed transaction in the last 90 days of the data, while 907 customers with money in the bank have never taken a loan.

### What we would do first

Add a second layer of checks on Mass Business and Mass Personal loans, where volume and default rates are both high. Set exposure limits for Home and Business lending and ask for a second sign-off on the largest loans. Ask operations to investigate failed transactions at Nagpur City and clear the pending backlog. Finally, make targeted loan offers to the 907 customers who already trust the bank with their money, and redefine what "Active" means for an account.

## 1. Loan defaults: which customers and products carry the risk?

***The question:*** *Among approved loans, which mix of customer segment and loan type defaults the most, in loan counts and in money?*

### What we found

Of 2,389 approved loans, 183 have defaulted. That is 7.7% of loans and about 170 million in value. By product, Education is the weakest at 11.1%, followed by Business (9.0%), Personal (7.3%), Auto (6.6%) and Home (6.2%).

Splitting by segment as well shows more. The highest default rates belong to Premium customers: 15.2% on Education loans (5 of 33) and 13.7% on Auto loans (7 of 51). Those groups are small, so a few loans moving either way would change the picture.

Most of the damage sits with Mass customers. They hold about 59% of approved loans but 120 of the 183 defaults, which is two thirds. Within Mass, default rates are highest on Business (10.7%) and Personal (8.5%) loans. In money terms, Mass Home and Mass Business loans lose the most, at about 41 million each. Mass Auto, at 7.1%, is close to the bank average and is not where the problem lies. Affluent customers are the safest group, with defaults as low as 3.1% on Auto and 4.7% on Home loans.

### What we suggest

- Add a second verification step, such as income proof and a check of existing debts, for Mass Business and Mass Personal applications. These have the most volume and the highest rates.
- Keep a close watch on Premium Education and Premium Auto, but collect a few more months of data before tightening policy, because the groups are small.
- Track defaults in money as well as percentages. Mass Home loans are large, so a small rise in their default rate would cost more than a big rise in Education.

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***How we looked at it:*** *Joined loans to customers, built the rates step by step with common table expressions, and used a window function to rank segments within each loan type. Only groups of at least 30 loans were compared.*

## 2. Loan approvals: do branches decide in the same way?

***The question:*** *How often does each branch approve loan applications, and which branches sit above or below the bank average?*

### What we found

The bank approved 74.7% of its 3,200 applications. Branch rates run from 78.6% at Hyderabad Central down to 67.5% at Kolkata Central. Nashik Road (78.0%) and Jaipur Main (77.9%) are also well above average. Bhubaneswar Main (68.4%), Nagpur City (71.3%) and Pune Main (71.5%) are the other branches below the line.

Kolkata Central and Bhubaneswar Main are also the two branches with the fewest applications, 114 and 98, so some of the gap could be chance. Customer type does not explain the differences either: Affluent customers are approved 75.4% of the time, Mass 74.4% and Premium 74.1%.

### What we suggest

- Review a sample of rejected applications at Kolkata Central and Bhubaneswar Main. The question to answer is whether they apply the same rules as the rest of the bank or are simply seeing weaker applicants.
- Share one approval checklist across all branches, so a similar applicant gets a similar answer wherever they apply.
- Do not treat the low rates as a fault yet. A low approval rate can be good risk control.

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***How we looked at it:*** *Counted approvals against all applications in each branch, then compared each branch with a bank-wide rate worked out from the totals, not by averaging branch rates. A window function placed the bank-wide figure next to every branch.*

## 3. Branch workload: how much does each employee handle?

***The question:*** *How much transaction value does each branch handle per active employee?*

### What we found

Ahmedabad Central handles about 8.0 million per active employee, with only four people on staff. Delhi Central is close behind at 7.8 million with six. At the other end, Bhubaneswar Main handles 1.2 million per employee, and Chennai Main and Nashik Road, with 13 and 11 staff, sit at about 1.5 million. The busiest branch carries roughly 6.5 times the load of the quietest.

Customers per employee tell the same story: 73 at Ahmedabad Central and 71 at Delhi Central, against 11 at Bhubaneswar Main. Value per employee is only a rough yardstick, because most transactions come from customers, not from staff effort, and the Ahmedabad figure rests on four people.

### What we suggest

- Check service levels and wait times at Ahmedabad Central and Delhi Central, and consider adding staff there.
- Before hiring, look at moving roles from the lightest branches, such as Chennai Main, Nashik Road and Bhubaneswar Main.
- Look at the mix of roles at each branch, not just the headcount.

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***How we looked at it:*** *Added up completed transaction value and counted active staff separately for each branch, then combined them. Doing it in that order stops the join from inflating the totals. Branches were then ranked.*

## 4. Seasonality and growth: is there a pattern over the year?

***The question:*** *Do transaction volumes follow a seasonal pattern, and how are we growing year on year?*

### What we found

Growth has been flat. Completed transactions were worth 106.9 million in 2023, 110.1 million in 2024 and 108.8 million in 2025, which is a rise of about 3% followed by a dip of about 1%. The best single month was December 2025 (10.2 million) and the weakest was June 2023 (8.0 million).

Averaging each calendar month across the three years, October is the strongest (about 9.75 million) and February the weakest (about 8.2 million). February is a shorter month, and it also has fewer

transactions (around 600 against 666 in October) and a smaller average transaction (13.7 thousand against 14.6 thousand). Still, the pattern is weak. Year-on-year changes for individual months swing from about -14% to +25% with no consistent direction, so most month-to-month movement looks like noise.

### What we suggest

- Do not build staffing or targets around strong seasonality. A light adjustment for a quieter February and a busier October is enough.
- Judge growth by quarter or year, not by single months.
- Flat volumes over three years mean growth will need active campaigns. It is not going to arrive on its own.

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***How we looked at it:*** *Grouped completed transactions by year and month, then used a window function to bring the same month of the previous year alongside each month and calculated the percentage change.*

## 5. Failed and pending transactions: where does money get stuck?

***The question:*** *Where do transactions fail or stay pending, and how much money is held up?*

### What we found

Out of 24,000 transactions, 869 failed or are still pending. That is 3.6%, and it leaves about 12.7 million stuck: roughly 5.4 million in failed transactions and 7.3 million in pending ones. The pending amount is larger, which suggests a lot of this money is waiting rather than lost.

By channel, the mobile app is the weakest at 4.4%, while net banking and debit cards are best at about 3.2%. The gaps between channels are small. The branch picture is clearer: Nagpur City stands out at 5.6% (71 failures out of 1,272 transactions), ahead of Hyderabad Central (4.2%) and Chennai Main (4.1%). The worst single combination is the mobile app at Nagpur City, where 15 of 160 transactions failed or stayed pending (9.4%). Nagpur's ATMs are also high at 5.6%. With that much volume behind it, the Nagpur gap looks real.

### What we suggest

- Ask operations and IT to look at the mobile app and ATMs at Nagpur City. Connectivity, ATM maintenance and settlement delays are the first things to check.
- Chase the 7.3 million in pending transactions, and set a rule for how long a transaction can stay pending before it is escalated.
- Track the failure rate monthly against the 3.6% bank average, and flag any branch that climbs well above it.

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***How we looked at it:*** *Measured failures against all transactions in each channel and branch (not just against other failures), compared them with the bank-wide rate and ranked them. Only groups with at least 30 transactions were included.*

## 6. Concentration risk: how much rests on a few big loans?

***The question:*** *Is the loan book concentrated in a few large loans, products or branches?*

### What we found

The bank has 1,692 active loans worth about 1.57 billion. The largest 10% of them (170 loans) hold 27.6% of that money, and the largest three tenths hold 61%. If loans were all the same size, the top 10% would hold 10%. Every one of those 170 big loans is a Home loan (133) or a Business loan (37). More than half of all Home exposure sits in those big loans.

Two products dominate the book: Home makes up 42.8% and Business 24.9%, which together is about 68%. Looking at branch and product together, 16 of 75 combinations hold half of the money and 38 hold 80%. The largest single combination, Home loans at Gurgaon Hub, holds 5.3% of the book, against an even share of about 1.3%. Delhi Central and Central Mumbai Home loans follow.

Some branches depend on big loans more than others. At Kolkata Central just over half of its lending comes from the bank's largest loans, though it has only 52 active loans. Central Mumbai (37%) and Gurgaon Hub (34%) come next, and Jaipur Main is lowest at 15%.

### What we suggest

- Set exposure limits by product and by branch, so no single area can grow unchecked.
- Require a second sign-off on the largest loans, which are all Home or Business.
- Run a stress test: what happens to the bank if the biggest Home loans at Gurgaon Hub, Delhi Central and Central Mumbai go wrong?
- Watch Kolkata Central closely. With so few loans, one default there would hurt.

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***How we looked at it:*** *Ranked active loans by size and split them into ten equal groups with a window function, then used running totals to show how quickly the cumulative share builds up. The same idea was repeated by branch and product.*

## 7. Growth opportunity: who could we offer a loan to?

***The question:*** *Which customers keep money with the bank but have no loan?*

### What we found

907 customers have an active account and no approved loan, and together they hold about 111.8 million. Mass is the biggest pool, with 537 customers and 65.5 million. Affluent has 280 customers and 35.3 million, and Premium has 90 customers and 11.0 million. Average balances are similar across the three, at about 122 to 126 thousand. The largest single balance belongs to an Affluent customer, at 1.35 million.

One gap to note: 743 of the 3,000 customers have no account records at all, so they could not be assessed.

### What we suggest

- Screen the list for eligibility, looking at income and credit history, then make targeted offers, starting with the top-ranked customers in each segment.

- Lead with safer products. For Mass customers, Auto and Home loans have had lower default rates than Business and Personal loans. Affluent customers, who default least, are the best place to grow lending.
- Find out why 743 customers have no account data. If they are real customers, they are part of the opportunity too.

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***How we looked at it:*** *Added up each customer's balances across accounts, used an anti-join to keep only customers with no approved loan, and ranked customers within each segment with a window function.*

## 8. Account activity: are "Active" accounts really active?

***The question:*** *How many accounts marked Active have had no completed transaction in the last 90 days of the data?*

### What we found

Of 3,630 accounts marked Active, 2,307 (63.6%) had no completed transaction in the last 90 days of the data, which ends in December 2025. Fifteen have never had one. Those accounts hold about 170.5 million of the 262.5 million sitting in Active accounts, or 65%. Current accounts are the most affected at 67.3%, with the other account types between 62.6% and 63.5%.

There is a caveat. Transactions in this dataset are sparse, roughly one per account every six months, so a 90-day window is strict and most accounts will look quiet. Even so, the Active label says very little about whether an account is really in use.

### What we suggest

- Define "active" by recent use, for example a transaction in the last six or twelve months, and not only by whether the account is open.
- Start a reactivation campaign with the highest-balance quiet accounts, since that is where most of the money is.
- Re-run the check with 180 and 365 days to see how much of the 63.6% remains before deciding how big a programme to run.

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***How we looked at it:*** *Found each account's latest completed transaction, joined it back to the Active accounts with a left join so accounts with no transactions are caught, and measured the gap using the last date in the data as the reference point, not today's date.*

## About the data and its limits

- The dataset is a public practice dataset and may be simulated. The findings describe this data, not a real institution.
- Default analysis uses approved loans only, since rejected and pending applications cannot default. "Exposure" means active loans.
- Where a finding rests on a small group, we have said so. Group comparisons used at least 30 loans or transactions.

- The data does not state a currency, so amounts are shown as plain figures. Loan and transaction amounts were stored as text and converted to numbers before analysis, which matters because text sorts incorrectly.

### Possible next steps

- Test default rates by loan size and interest rate to see what drives the risk inside each segment.
- Repeat the failure and approval checks every quarter and track whether the suggested changes make a difference.
- Repeat the account activity check with longer windows.